

BACK PAPER

Elementary Number Theory

Instructor: Ramdin Mawia

Marks: 45

Time: December 27, 2024; 10:00–13:00.

Attempt FIVE problems. The maximum you can score is 45.

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1. Let a and b be relatively prime positive integers. For any integer $c > ab - a - b$, prove that there are nonnegative integers u and v such that $au + bv = c$. Also show that $ab - a - b$ cannot be expressed as a linear combination of a and b with nonnegative coefficients. **10**
 2. Compute the class number $h(-52)$. **10**
 3. Prove that there are infinitely many primes of the form $8k - 1$. [Hint. Assume $p_1, \dots, p_n$ are all the primes of the form $8k - 1$. Look at $(p_1 \cdots p_n)^2 - 2$ and apply QRL.] **10**
 4. Solve the congruence $3X^2 + 11Y^2 \equiv 13 \pmod{47}$ over $\mathbb{Z}$. **10**
 5. Let $(u_n)_{n \geq 0}$ be a complex sequence satisfying a linear recurrence of order 2. Suppose $u_0 = 0, u_1 = 1, u_2 = 3, u_3 = 4$. Find the n th term u_n of the sequence in closed form. **10**
 6. Prove that the sum of the reciprocals of the primes diverges. **10**
 7. Given that $p = 2^{2^4} + 1 = 65537$ is a prime ("Fermat prime"), prove that $a \in \mathbb{Z}$ is a primitive root mod p if and only if it is a quadratic nonresidue mod p . Use it to find the period¹ of the rational number $1/65537$. **10**

¹The *period* of a rational number is the number of repeating digits in its decimal expansion. Eg. $1/7 = 0.\overline{142857}$ has period 6.